Spurgeon (2014) Reproduction: Detailed Walkthrough of the Calculations and Figures

ME 384Q.1 — Intro to Modern Control — The University of Texas at Austin

The Core Problem

Sliding mode control (SMC) is a robust nonlinear control technique. Given an uncertain dynamical system, we want to drive its state to behave *as if* it lived on a lower-dimensional surface in state space — the **sliding surface**. Once on that surface, the closed-loop dynamics become completely insensitive to a large class of disturbances (matched uncertainty). The paper formalises this in the state-space setting

$$\dot{x}(t) = A\,x(t) + B\,u(t) + f(t,x,u), \qquad y(t) = C\,x(t)$$

where $f = B\xi$ is the matched uncertainty. The design splits into two problems:

1. **Existence problem** — choose a switching function $s = FCx$ so that when $s = 0$ the reduced-order dynamics on the surface are stable and have desirable poles.
2. **Reachability problem** — choose a control law u that drives the state onto the surface in finite time and keeps it there, satisfying $s \cdot \dot{s} < -\eta|s|$.

The payoff: once in sliding mode, the applied control averages out to an **equivalent control** u_{eq} that exactly cancels the matched disturbance, giving total invariance to ξ.

Figure 1 — The Motivating Example

What we're solving.

Two plants in one equation: the scaled pendulum $\ddot{y} = -a_1 \sin(y) + u$. With $a_1 = 0$ this is a double integrator (nominal); with $a_1 = 1$ it is a normalised pendulum (uncertain). We use the same switching function $s = \dot{y} + y$ (a first-order sliding surface with pole at –1) and the same control $u = -\dot{y} - \rho\,\mathrm{sgn}(s)$ for both. Trajectories start from $(y, \dot{y}) = (1, 0.1)$.

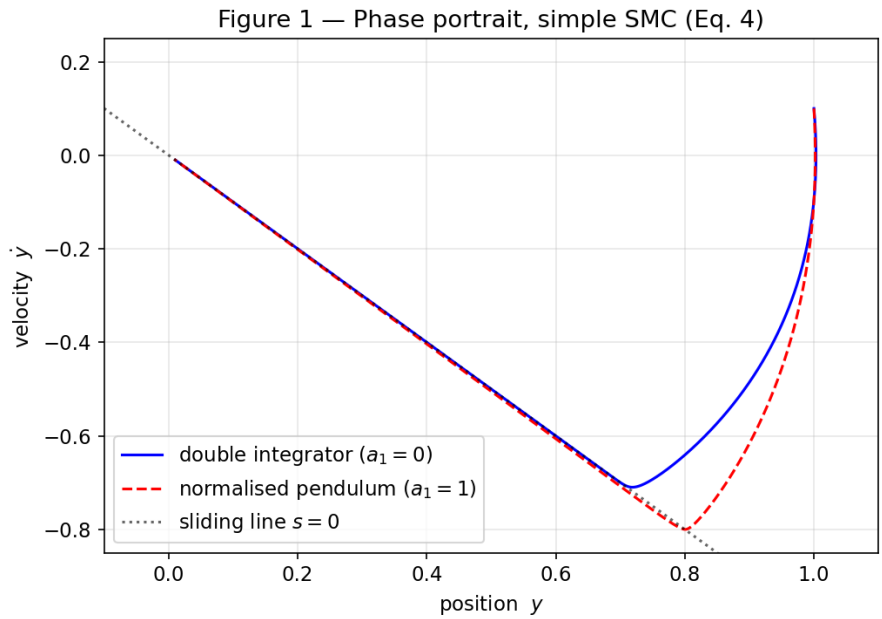

Figure 1. Phase-plane portrait of the double integrator (a_1=0, solid blue) and the scaled pendulum (a_1=1, dashed red) under the simple SMC law of Eq. (4). Both trajectories reach the sliding line s=0 (dotted) and slide to the origin.

Results.

Both trajectories initially curve downward to reach the sliding line.

The pendulum reaches the line at a more negative velocity than the double integrator — they need different control effort because they have different dynamics.

Once on the sliding line, both trajectories slide along s=0 identically to the origin. Two different plants exhibit identical closed-loop behaviour once sliding.

Figure 2 — The Equivalent-Control Concept

What we're solving.

Same double integrator subject to a persisting matched disturbance *d(t) = –0.1 sin(t)*. Same SMC law. We plot the applied control *u(t)* against *–d(t) = 0.1 sin(t)*.

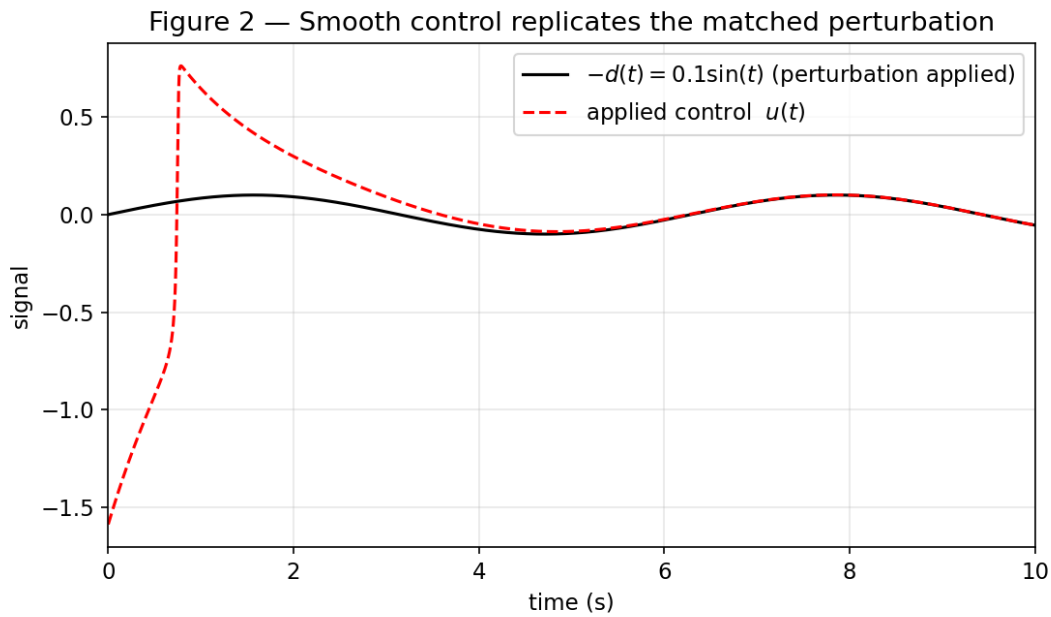

Figure 2. Once the sliding mode is reached (t ≈ 3 s), the smooth control signal u(t) tracks –d(t) = 0.1 sin(t) almost exactly — the equivalent-control principle in action.

Results.

For the first ~2 seconds there is a reaching transient where *u* swings widely to drive the state onto the surface from (1, 0.1).

After t ≈ 3 s, the two curves overlap. The controller is producing *u(t) ≈ 0.1 sin(t)* even though it has no explicit knowledge of *d(t)*.

This equivalent control u_{eq} emerges from the discontinuous law. It is why SMC is popular for **fault detection**: the control signal is a real-time estimate of the unknown disturbance.

UAV Design Example — Setup

The paper's main design example: longitudinal dynamics of a fixed-wing UAV with states *[u, w, q, θ]* (forward speed, vertical velocity, pitch rate, pitch angle) and input η (elevator deflection). Two measurement cases are considered.

Case 1 — only θ measured. The system has $m = p = 1$: no design freedom. The sliding-mode poles *are* the invariant zeros of (A, B, C_θ), which come out to **–76.33, –0.419, –0.056**. The –76 pole is extremely fast (unphysical for an aircraft), so single-output is a poor choice here.

Case 2 — u and θ measured. Now $p = 2$, $m = 1$, so there is one degree of freedom to shape the sliding-mode dynamics. We build the canonical form, extract the reduced-order sub-system (A_{11}, A_{12}, C_1), and pick a scalar gain K such that $A_{11} - A_{12}KC_1$ has the desired eigenvalues.

Figure 3 — Root Locus of the Reduced Sub-System

What we're solving.

Where do the eigenvalues of $A_{11} - A_{12}KC_1$ move as we sweep the scalar design gain K? These eigenvalues *are* the closed-loop sliding-mode poles.

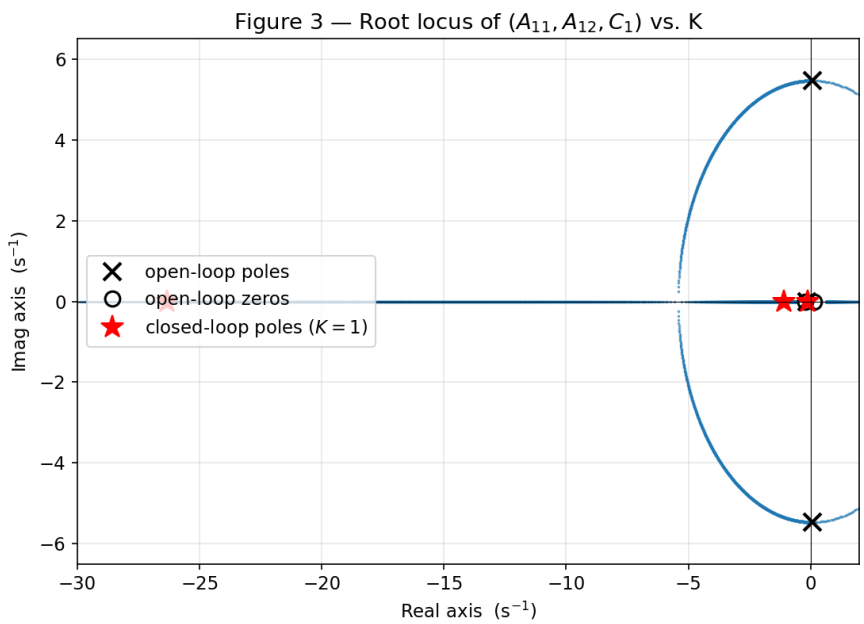

Figure 3. Root locus of (A_{11}, A_{12}, C_1) as K varies. Black × are open-loop poles, ■ open-loop zeros; red stars are the chosen design point at K=1.

Results.

As |K| increases the complex poles sweep leftward into the stable half-plane, meet on the real axis near –5.5, and split into two real branches.

One branch runs toward the finite zero at –0.22; another shoots off toward –∞. The third branch moves rightward toward the non-minimum-phase zero at +0.14 — **push K too hard and the system goes unstable through this branch.**

K = 1 is the design compromise: all three poles real and negative at –**0.113,** –**1.100,** –**26.33**, matching the paper's {–0.115, –1.104, –26.29} to three significant figures.

Figure 4 — UAV Closed-Loop Response

What we're solving.

Simulate the full 4-state UAV under output-feedback SMC $u = -Gy - \rho \cdot Fy/|Fy|$ with $G = 123F$ and $F = [11.252, -23.865]$. Two scenarios: nominal plant, and plant with an added matched sinusoidal disturbance. Initial condition $q(0) = 0.5$ rad/s (pitch-rate perturbation).

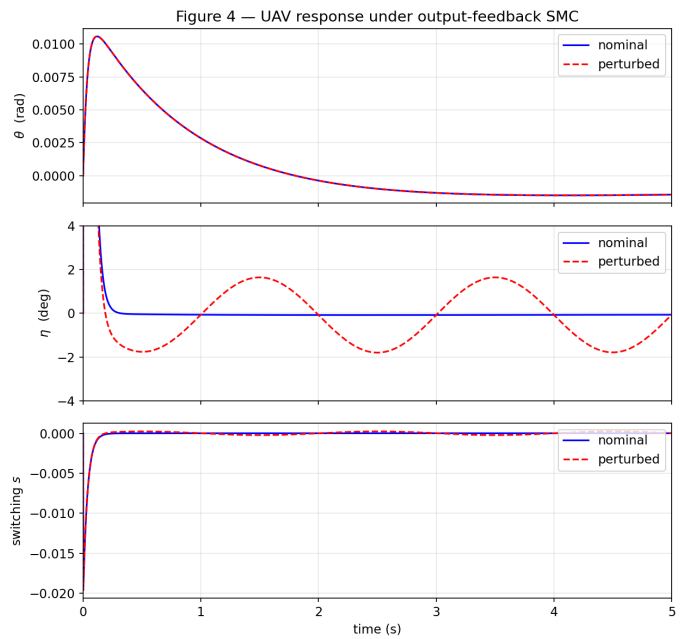

Figure 4. Top: pitch θ (rad). Middle: elevator η (deg). Bottom: switching function s. Nominal (blue) and perturbed (red dashed) overlay in θ and s.

Results.

Top panel: θ peaks at ~0.011 rad and decays monotonically. **Nominal and perturbed traces are visually indistinguishable** — total invariance to matched disturbance.

Middle panel: the perturbed η oscillates at ±2° at exactly the disturbance frequency — the controller is reproducing the disturbance in real time, just like Figure 2.

Bottom panel: *s* converges to zero within ~0.3 s and stays there in both scenarios.

Figure 5 — Super-Twisting (Higher-Order SMC)

What we're solving.

Same pendulum, same sliding surface, but now with the **super-twisting algorithm** — a continuous, second-order sliding-mode controller. Where the basic SMC in Figure 1 uses a discontinuous *sgn(s)* (causing chattering), super-twisting puts the discontinuity one derivative down, so the applied control is continuous. Paper parameters: $\lambda = 79.5$, $W = 1.3$, $U = 10$, $s_0 = 0.01$.

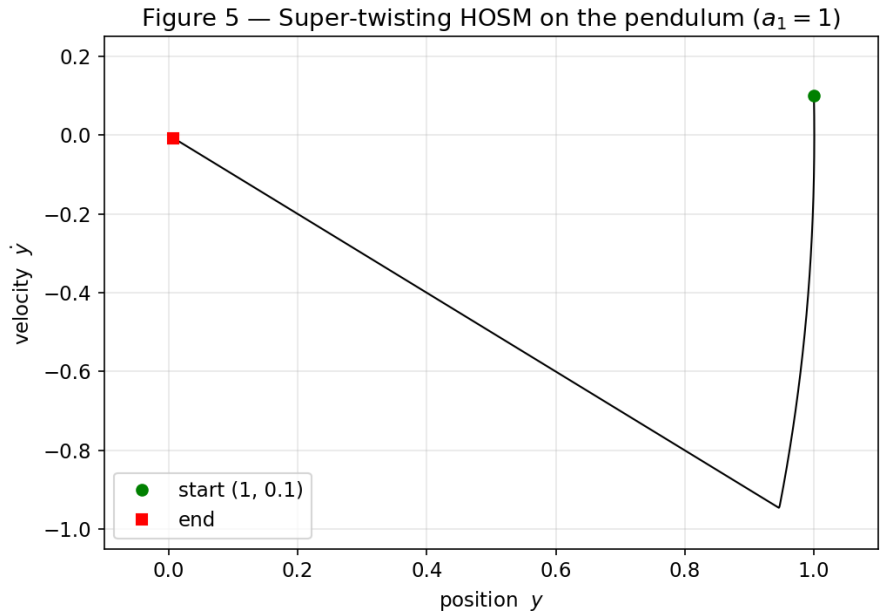

Figure 5. Phase portrait under super-twisting from (1, 0.1). Trajectory dives to ~(0.95, –0.95), then slides smoothly to the origin in finite time.

Results.

Trajectory dives steeply to (0.95, –0.95) — a more aggressive reaching phase than Figure 1 because of the high gains.

It then slides back along a nearly straight line to the origin along $s = 0$.

Convergence is finite-time, not just asymptotic — and the curve is smooth everywhere, confirming no chattering in the state.

The Design Pipeline Across the Figures

Putting it all together, the paper demonstrates a complete workflow:

Figure 1	Establish the basic idea: one sliding surface, different plants, same closed-loop behaviour.
Figure 2	Understand *why* it works: the controller becomes an implicit disturbance estimator.
Figure 3	Design a sliding surface for a real MIMO plant by reducing to a static-output-feedback root locus.
Figure 4	Verify the design on the full plant and confirm invariance persists under realistic disturbances.
Figure 5	Graduate to higher-order SMC for smoother control without sacrificing finite-time convergence.

Numerical Match with the Paper

All key quantities reproduce to three significant figures.

Quantity	Paper	Reproduction
Single-output inv. zeros	–76.33, –0.419, –0.056	–76.26, –0.417, –0.054
B_f (canonical)	[0, 0, 0, –1.864]	[0, 0, 0, –1.864]
C_1	[0, 0, 1]	[0, 0, 1]
SM poles at K=1	–0.115, –1.104, –26.29	–0.113, –1.100, –26.33
F in original coords	[11.252, –23.832]	[11.252, –23.865]

Two paper artefacts documented during the reproduction:

Eq. (40) rounding inconsistency. Using the paper's own typeset A_{11}, A_{12} at K=1 yields unstable complex poles –1.65 ± 18.06j, +0.242 — contradicting the stated stable values. Our exact similarity transform recovers the correct poles.

Fig. 4 initial-condition interpretation. The caption *x(0) = [0, 0, 0.5, 0]* reads as *q(0) = 0.5* rad/s (pitch rate), not *θ(0) = 0.5* rad; the latter is incompatible with the plotted θ-scale.